

1. Administrative & Study Identification

Full Study Title: A Safety and Efficacy Study of Anti-inflammatory (Canakinumab) and Cartilage Stimulating (LNA043) Drugs Injected Into the Knee Joint of Participants With Knee Osteoarthritis (OA) **Short Title/Acronym:** CLNA043A12203 **Protocol Number:** CLNA043A12203 **NCT Number:** NCT04814368 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Canakinumab (ACZ885; Trade Name: Ilaris, ATC: L04AC08) and LNA043 (ANGPTL3 derivative) **Phase of Development:** Phase II **Medical Monitor:** Novartis Medical Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the safety, tolerability, and efficacy (specifically the change in Cartilage Volume measured by MRI and the change in Knee Injury and Osteoarthritis Outcome Score [KOOS] Pain Subscale) of intra-articular canakinumab followed by intra-articular LNA043 in participants with symptomatic knee osteoarthritis. * **Secondary Objectives:** To assess immunogenicity (anti-LNA043 antibodies), and maximum systemic and synovial fluid concentrations of ANGPTL3.

2.2 Background & Rationale To address the critical unmet medical need for a Disease-Modifying Osteoarthritis Drug (DMOAD) that provides both symptomatic pain relief and structural joint repair. OA synovial inflammation is heavily driven by pro-inflammatory cytokines like IL-1 β , which sensitizes pain pathways. Canakinumab acts to block IL-1 β , diminishing joint inflammation. In tandem, LNA043 is utilized to stimulate endogenous chondrogenesis and regenerate hyaline cartilage matrix.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional (Non-confirmatory) * **Control Type:** Placebo-controlled * **Blinding:** Quadruple (Participant, Care Provider, Investigator, and Sponsor-blinded) * **Randomization:** Randomized, Four-arm parallel group

3.2 Planned Enrollment & Duration * **Target \$N\$:** 23 participants targeted and enrolled (study was terminated early). * **Number of Sites:** Approximately 11 study locations. * **Estimated Study Start:** 27-Aug-2021 * **Trial Status:** Terminated (Study was terminated early)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

1. Moderate to severe OA pain (corresponding to NRS Pain ≥ 5 to ≤ 9) in the target knee for the majority of days in the last 3 months prior to Screening.
2. KOOS pain subscale < 60 for the target knee during Screening.
3. High sensitivity C-reactive Protein (hsCRP) ≥ 1.8 mg/L.
4. Radiographic KL grade 2 to 4 knee OA and joint space width (JSW) 2.0-4.0 mm (men) or 1.5-3.5 mm (women) in the medial tibiofemoral compartment (TFC) in the target knee.
5. Contrast-enhanced MRI (CE-MRI) diagnosed moderate or severe knee synovitis based on an established synovitis scoring system (moderate score 9-12 or severe score ≥ 13).

4.2 Exclusion Criteria

1. History of, or planned; knee replacement (partial or total) in either knee; arthroscopy or lavage in either knee within 6 months prior to screening; any other previous surgical intervention in the target knee, or for the contralateral knee within 12 months prior to Screening, including mosaicplasty, microfracture, meniscectomy $>50\%$ or osteotomy.
2. Moderate to severe pain in the contralateral knee for the majority of days in the last 3 months prior to screening.
3. Malalignment $> 7.5^\circ$ in the target knee (either varus or valgus).
4. Any diagnosis of systemic inflammatory arthritis or connective tissue disease, including but not limited to rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, gout, Paget's disease, systemic lupus erythematosus) or other systemic condition that might confound assessment of OA (e.g. fibromyalgia).
5. Ipsilateral hip OA or hip prosthesis recently implanted (within 1 year prior to screening) or hip replacement on either side planned within the study period.

5. Intervention & Treatment Plan

5.1 Investigational Regimen

- * **Dosage/Frequency:** Intra-articular (IA) injections of Canakinumab followed by IA LNA043 (or their matching placebos) dependent on the assigned study arm.
- * **Route of Administration:** Intra-articular (IA) injection.
- * **Duration of Treatment:** Targeted endpoint analysis up to Day 197 with continued monitoring.

5.2 Concomitant & Prohibited Therapy

- * **Permitted:** Standard baseline analgesic care as defined by protocol.
- * **Prohibited:** Concurrent DMOADs, unauthorized biological therapies, or alternative IA injections (steroids/hyaluronic acid) during the study protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Prior to Day 0	Informed Consent, CE-MRI, X-ray, Labs, KOOS, and NRS Pain assessment
Baseline	Day 0	Randomization, First IA Injection
Follow-Up	Day 197	Primary Efficacy Assessment (Cartilage Volume MRI), KOOS questionnaire evaluation
End of Study	Upon Early Termination	Final safety assessments, AE reporting, Immunogenicity labs

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. Change in Cartilage Volume in the Index Region Measured by MRI at Day 197 (Placebo to ACZ885+LNA043 Versus Placebo to ACZ885):

Magnetic resonance images (MRI) obtained from the target knee to visualize and quantify changes in volume of cartilage in the index region. The index region was defined as the union of the femoral medial anterior (FMA), central (FMC) and posterior (FMP) cartilage subregions in the knee. 2. Change in Knee Injury and Osteoarthritis Outcome Score (KOOS) Pain Subscale (ACZ885 Versus Placebo to ACZ885):

Assessment of patient's perception about their knee pain. The KOOS pain subscale consists of 9 questions with a recall of 7 days. Each question has 5 standardized answer options (scored 0 to 4). A normalized score is calculated (100 indicating no symptoms and 0 indicating extreme symptoms).

7.2 Secondary Endpoints 1. Number of Participants With Anti-LNA043 Antibodies (Placebo to ACZ885+LNA043 Versus ACZ885 + LNA043). 2. Maximum Serum Concentration (C_{max}) of ANGPTL3 (Placebo to ACZ885+LNA043 Versus ACZ885+LNA043). 3. Synovial Fluid Concentrations of ANGPTL3 (Placebo to ACZ885+LNA043 Versus ACZ885+LNA043).

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous collection of AEs and SAEs directly related to the intra-articular injections.
 - **Serious Adverse Events (SAEs):** Evaluated strictly through sponsor pharmacovigilance up to study termination.
 - **Data Monitoring Committee (DMC):** Supervised and governed by Novartis Pharmaceuticals safety review panels.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Final evaluation population adapted based on early termination.

- **Sample Size Justification:** A sample size calculation was originally planned to provide 80% power ($\alpha = 0.05$) to detect significant differences in structural and symptomatic parameters (MRI cartilage volume and KOOS) between treatment groups.
 - **Handling of Missing Data:** Due to significant recruitment challenges and early termination of the study (status: TERMINATED, 23 participants total enrolled), inferential statistics were restricted; analyses default to a descriptive synoptic clinical study report.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Supervised centrally by Novartis with independent review of imaging endpoints.
 - **Audit Trail:** Detailed tracking and eCRF compliance monitored through standardized trial management software.
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11. Study Identifiers & Governance

- **Primary ID:** CLNA043A12203
 - **Registry Number:** NCT04814368
 - **Sponsor/Lead Organization:** Novartis Pharmaceuticals
 - **Study Title:** A Safety and Efficacy Study of Anti-inflammatory (Canakinumab) and Cartilage Stimulating (LNA043) Drugs Injected Into the Knee Joint of Participants With Knee Osteoarthritis (OA)
 - **Official Regulatory Title:** A Randomized, Four-arm, Canakinumab Placebo-controlled, Participant, Investigator and Sponsor-blinded Study Investigating the Safety, Tolerability and Efficacy of Intra-articular Canakinumab Followed by Intra-articular LNA043 in Patients With Knee Osteoarthritis
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12. Clinical Framework (Adapted for Knee OA Protocol)

- **Primary Condition:** Knee Osteoarthritis (OA) with active synovial inflammation
 - **Diagnosis Threshold:** KL grade 2-4, medial TFC JSW threshold, and hsCRP ≥ 1.8 mg/L
 - **Medical Methodology:** Intra-articular Disease-Modifying Osteoarthritis Drug (DMOAD) Intervention
 - **Optimization Target:** Inflammatory pain reduction and targeted hyaline cartilage regeneration
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13. Study Procedures & Methodology

- **Management Pathway:** Sequential IA biological injections utilizing an anti-inflammatory followed by a chondrogenic agent
 - **Education Protocol (HCP):** Standardization of CE-MRI Ktrans synovitis scoring and IA injection administration
 - **Education Protocol (Patient):** Guidance on accurate 7-day recall for the KOOS pain subscale evaluations
 - **Quality Audit Mechanism:** Descriptive tracking of early termination data and protocol deviations including COVID-19 related impacts
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14. Observation & Follow-Up Schedule

- **Total Duration:** Scheduled observation up to Day 197 endpoint; terminated early
 - **Onsite Visit Cadence:** Baseline, subsequent IA injections, and primary assessment evaluation
 - **Remote Monitoring Frequency:** Adverse event and symptomatic monitoring as required
 - **Checklist Review Interval:** Data lock triggered prematurely due to early study termination
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15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				